printf("enter two integer numbers")
→ scanf("%d, %d", &num1, &num2);

void main

{

① - &

}

printf();

not, if;

Q. Write a p
your deprtn

#include <st
void main();
{
printf("tha
}

→ we can w
but under c

/n → for outp

/t → for spa

Q. Write a program to find the avg of any three integer numbers.

```
#include <stdio.h>
void main()
{
int num1, num2, num3, avg;
printf("enter two integer numbers");
scanf("%d, %d, %d", &num1, &num2, &num3);
avg = (num1 + num2 + num3)/3;
printf("The average of the numbers ~~are~~ is %d", avg);
}
```

enter two integer numbers : 2, 5, 3

The average is 3

10
10

ASCII : American standard code for Information Interchange.

character encoding standard that assign unique numerical values to characters, including letters, digits, punctuation marks and control characters.

6<=5 False - loop stops

2) For loop:

Memory

N	sum	i
5	0	1
3987	4051	4096

K<=5 True

	sum	i
5	1	2

2<=5 True.

3	3

3<=5 True

6	4

4<=5 True

10	5

5<=5 True

15	6

6<=5 False:

loop stops.

output:

enter the limit 5.

{1 + 2 + 3 + 4 + 5}

The sum of first 5 natural nos: is 15.

10

10

Q. Write a program to find the factorial of ~~an~~ a number.

2. print the reverse of a number:

3. Palindrome

4. Fibonaccio series: 0, 1, 1, 2, 3, 5, 8, 13....

the opshn to read and display a 2x2 matrix :

```
cludecstdio.h>
main()

int mat[100][100], i, j, n, m;
printf("Enter the limit for matrix");
scanf("%d %d", &m, &n);
printf("enter the matrix elements");
for(i=0; i<m; i++)
{
  for(j=0; j<n; j++)
  {
    scanf("%d", &mat[i][j]);
  }
}
for (i=0; i<m; i++)
{
  for(j=0; j<m; j++)
  {
    printf("%d\t", mat[i][j]);
  }
  print("\n");
}
```

10

else
sub3 is larger.

else ~~sub~~ if (sub2 > sub3)
printf sub 2 is larger.

else
sub 3 is larger.

Q. WAP to swap two numbers using fnctn w/o return type and ~~and~~ arguments.

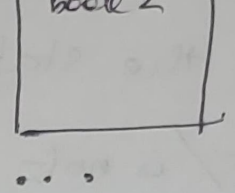
2) w/o return type and with arguments.

→ pass by value method
→ call by value.

the opns occuring during copy do not affect the ~~fnctn~~ actual argument.

only copy of actual argumnt is passed ~~during~~ to fnctn def.

storing structure consecutive place.



Q. Write a programme to declare a structure with ~~name~~ structure name Marks,
struct members are : an integer value total mark.
and an integer array of subject which can store mark of 5 subjects.
structure variable is an array of structure called student.
that can store the details of three students.

WAP To calculate the total marks of all subjects of each student. Also calculate the total marks gained by each student for each subject.

5 m
10

p1 : Start

p 2 : Initialize the process and define variable PI 3.14159 and introduce header file <math.h>

p 3 : Declare function Diameter (int x_1, int x_2, int y_1, int y_2)

p 4 : define the function and calculate diameter $D = sqrt((x_2 - x_1)^2 - (y_2 - y_1)^2)$

p 5 : Declare function Area (int r)

p 6 : Define function and calculate Area, A = PI * R * R.

ep 7 : Declare function circumference (int P)

p 8 : Define function and calculate P = 2 * PI * R.

tep 9 : Read x_1, x_2, y_1, y_2 and r

tep 10 : Call functions Diameter (x_1, x_2, y_1, y_2)

tep 11 : Call function Area (r)

step 12 : Call function circumference (r)

step 13 : return 0

step 14 : Stop.

2/22.

Diameter (int a1, int a2, int b1, int b2)

step 1 : ~~initialize~~ declare variable D;

step 2 : D = sqrt (pow((a2 - a1), 2) + pow(b2 - b1), 2))

step 3 : print diameter D

step 4 : stop

Area (int R)

Step 1 : declare variable A

Step 2 : Area, A = PI * R * R

Step 3 : print the " Area of circle = A.

Step 4 : stop.

2. step1 : Start

step 2 : Declare variable sub1 , sub2, sub3

step 3 : ~~display~~, read the value for sub1, sub2, sub 3 from user.

step 4 : Call Marks(sub1, sub2, sub3)

step 5 : call smallest (sub1, sub2, sub 3 .

step 6 : call Average (sub1, sub2, sub3)

step 7 : Stop.

12.
2e

Marks(int s1, int s2, int s3)

step1 : Start .

step 2 : if(s1 == s2 && s2 == s3) then display .
" all the subjects, marks are same". End IF .

step 3 : if (s1 > s2) Then . go to step 4 , if not go to step 8.

step 4 : if (s3 > s1) then goto step 5 . if not go to step .

step 5 : if (s3 > s2) , ~~check~~ then check if s3 == s2, if not .
go to step6, if not goto step .

step 6 : display "subject3 has the maximum marks scored".

step 7 : s3 == s1, subject and subject scored the same marks .

step 8 : else display subject 1 has the maximum score . End IF .

step 9 : ~~else if~~ otherwise (s2 >= s3) then go to step 10 .

step 10 : check if (s2 == s3) , then print subject2 and subject 3 .
have the same marks and are the maximum if not go to .

step 11 : else display "subject 2 has the maximum marks .

step 4: result = sum.

step 5: display "cos(x) = " result.

step 6: Return.

SUM ()

step 1 : Start

step 2 : initialize float sum = 0, term = 1, result :
int n = 1.

step 3 : while (n==1 || (term/sum) > 0.000001)) Do:

term = pow((1/(factorial)n, n));

sum = sum + term;

n++

End while.

step 4 : result = sum.

step 5 : Display " SUM = " result.

step 6 : Return.

$\frac{|2}{2}$

4.

step1: start.
step2: Define variable MAX. is 10
step3: Declare variables: m, n, i, j and matrix mat [MAX] [MAX]

step 4: Read m, n.

Step 5: Print "Enter matrix elements".

step 6: for (i = 0; i<m ; i++) Do.
for j=0, j<n, j++. Do.
Read mat [i] [j]
End For
End For.

step 7: Call findsaddlePoints (mat, m, n)
step 8: Call findSecondLargest (mat, m, n)
step 9: Stop

1/22

findSaddlePoints (int mat, m, n)

step1: Start.
step2: Declare variable i, j, k, found = 0.
step 3: FOR i = 0 TO m-1 DO - go to step 4.
step 4: Set int min = mat[i] [0], col_index = 0
step5: IF mat [i] [j] < min THEN
min = mat [i] [j]
col_index = j.
END IF
END FOR.

step6: set int is_saddle = 1
step 7: FOR k = 0 to k = m - 1 DO

~~Rep~~ readData(s, n);

Step 1 : Start

step 2 : For i=0 to n-1 Do.

print "enter details for student i+1"

Print "Roll no:"

~~red~~ read s[i].roll

Print " Name"

read s[i].name

Print "Enter Total marks:

read s[i].total marks.

End FOR.

Step3: Return.

printlist(s,n):

step1: Start.

step2: Print "Rank", "Roll", "Name", "Total Marks".

step 3: For i =0 to n-1 Do.

print i+1, s[i].roll, s[i].name, s[i].totalmarks.

END FOR.

step 4. Return.